SCTR's Pune Institute of Computer Technology, Pune - 411043

Department of Electronics & Telecommunication Engineering

Division: SY- VIII **Batch: G8**

Subject: Project Based Learning (PBL)

Topic:

IoT-Enabled Smart Stick for Visually Impaired People

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INTRODUCTION

Visual impairment significantly affects an individual's ability to move safely and independently. Traditional white canes provide limited support as they rely only on physical contact with nearby objects.

Advancements in embedded systems and assistive technology have enabled the development of smart devices that improve obstacle awareness and user safety. Such systems can detect hazards in advance and provide timely alerts.

This project aims to design an intelligent assistive stick that enhances mobility by offering real-time obstacle detection and feedback, thereby improving confidence, safety, and independence for visually impaired individuals.

LITERATURE REVIEW

Paper 1: Smart Walking Stick for the Visually Impaired Using Ultrasonic Sensors

Authors	Publication	Proposed Work	Outcomes	Design Methodology
Gbenga D.E. Shani A.I. Adekunle A.L.	**International Journal of Engineering & Technology**	This paper presents a smart walking stick designed using an ultrasonic sensor to detect obstacles ahead of the user. It provides simple distance-based alerts to improve navigation for visually impaired individuals.	The system successfully detects nearby obstacles and alerts the user using a buzzer, improving basic mobility and safety compared to a traditional white cane.	✓ Ultrasonic Sensor ✓ Arduino Microcontroller ✓ Buzzer for Audio Alerts ✓ Simple Threshold-Based Distance Measurement
		The design focuses on using a single sensor and basic control logic to create an affordable assistive device.	Improved awareness of front obstacles in indoor and outdoor environments.	✓ Low-Cost Hardware ✓ Embedded System Design ✓ Basic Programming (C/C++)
		The aim is to provide a cost-effective solution for visually impaired individuals.	Helps users avoid frontal obstacles, reducing collision risks.	✓ Battery-Powered Portable System
Research Gaps / Limitations	The system offers only basic front obstacle detection and lacks water, overhead, and side sensing. It also has no camera, IoT connectivity, vibration feedback, or emergency communication features included in our proposed ESP32-CAM design.			

LITERATURE REVIEW

Paper 2: Assistive Infrared Sensor–Based Smart Stick for Blind People

Authors	Publication	Proposed Work	Outcomes	Design Methodology
A.A. Nada M.A. Fakhr A.F. Seddik	IEEE Science and Information Conference (SAI) 2015	This paper proposes a smart stick system using infrared (IR) sensors to detect nearby obstacles and stair edges. The system alerts the user using a buzzer when an obstacle is detected at a predefined distance.	The system offers effective short-range obstacle detection and staircase identification, making mobility safer for visually impaired users in controlled environments.	✓ Infrared (IR) Sensors ✓ Microcontroller-Based Control ✓ Buzzer for Audio Alerts ✓ Simple Threshold Detection Method
		The design emphasizes a lightweight and low-cost structure suitable for daily use by blind individuals.	Provides quick response to close obstacles with minimal hardware complexity.	✓ Low-Cost Embedded Hardware ✓ Compact, Portable Stick Design
		The work focuses on simple and affordable electronic components for ease of deployment and maintenance.	Helps users navigate indoors with basic hazard detection.	✓ Battery-Powered System ✓ Basic Embedded Programming
Research Gaps / Limitations	IR sensors have short detection range and are affected by ambient light, leading to unreliable sensing in outdoor environments. The system lacks water detection, overhead sensing, IoT connectivity, camera support, and emergency communication features present in our ESP32-CAM design.			

LITERATURE REVIEW

Paper 3: An Intelligent Walking Stick for the Blind Using Ultrasonic & Voice Feedback

Authors	Publication	Proposed Work	Outcomes	Design Methodology
S. Ulrich **M. C. Domingo** **J. Sani**	**IEEE International Conference on Robotics and Automation (ICRA) 2014**	This paper presents an intelligent walking stick that uses ultrasonic sensors to detect obstacles and provides voice-based guidance to the user. The aim is to improve mobility by offering directional information through audio feedback.	The system accurately detects obstacles in the user's path and delivers real-time audio instructions, helping visually impaired individuals move more confidently and avoid collisions.	✓ Ultrasonic Sensors ✓ Microcontroller-Based Processing ✓ Voice Output Module (Audio Guidance) ✓ Threshold-Based Obstacle Detection
		Focuses on enhancing user safety by integrating sound alerts that communicate obstacle distance and direction.	Enhanced navigation with intuitive voice instructions instead of beeps or vibration alone.	✓ Embedded System Design ✓ Portable, Battery-Powered Architecture
		Designed as a low-cost smart assistive device with easy-to-use auditory interface.	Provides effective indoor navigation support with minimal hardware complexity.	✓ Basic Programming (C/C++) ✓ Small Form Factor Stick
Research Gaps / Limitations	The system relies solely on ultrasonic sensing and voice alerts, lacking water detection, overhead sensing, IoT tracking, and camera-based visibility. It offers limited functionality compared to our more advanced ESP32-CAM multi-sensor design.			

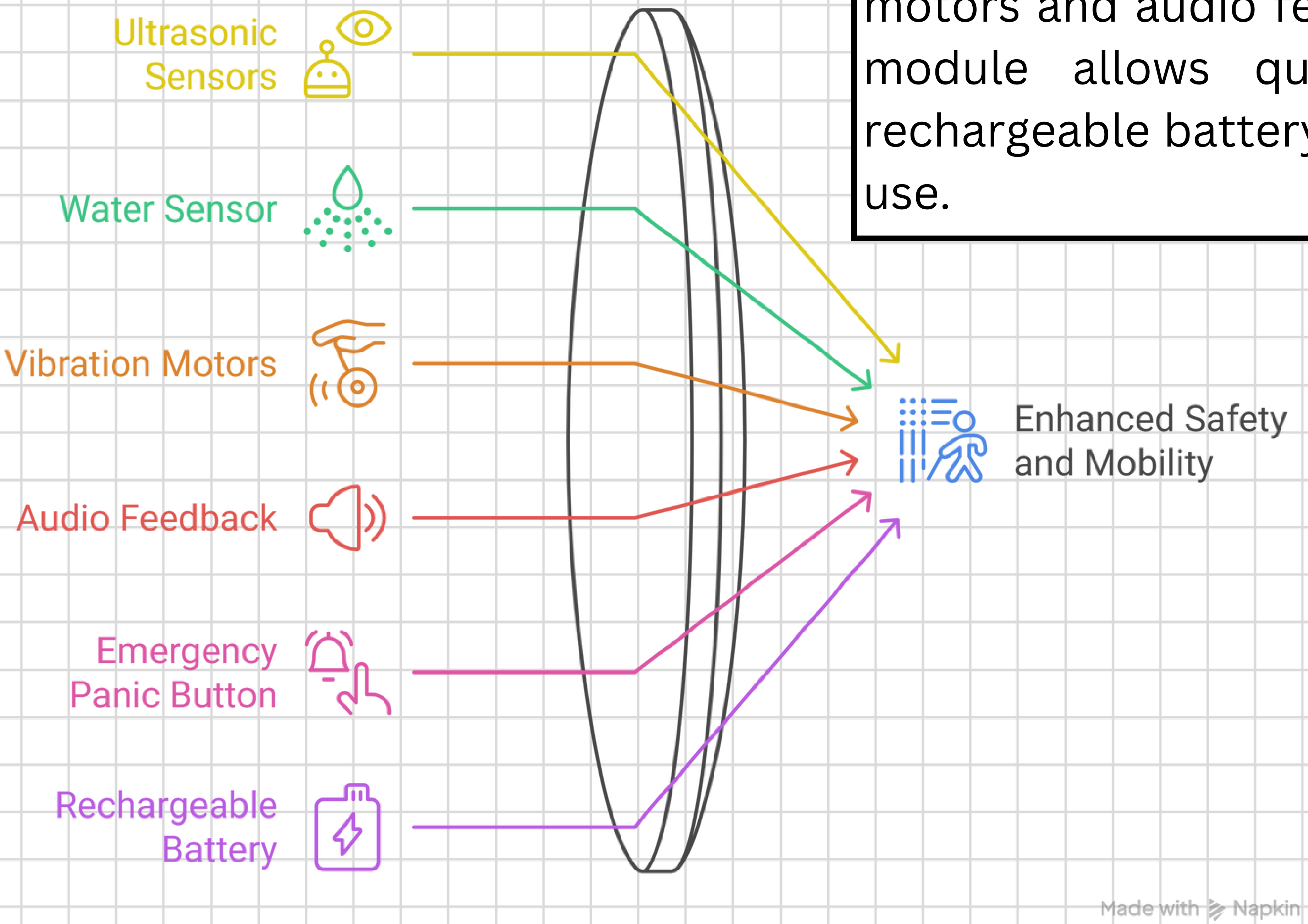
LITERATURE REVIEW

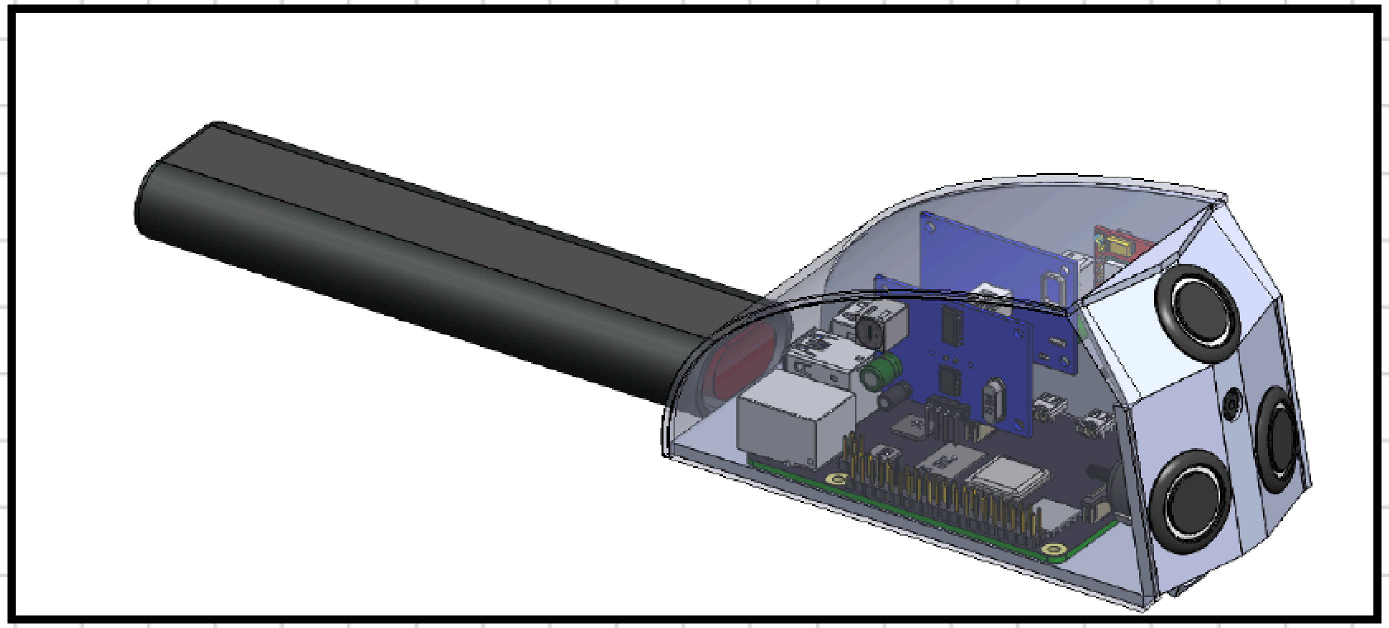
Paper 4: A Smart Assistive Navigation System for the Visually Impaired Using Multi-Sensor Fusion

Authors	Publication	Proposed Work	Outcomes	Design Methodology
K. Bhatia R. Sharma P. Kumar	IEEE Access 2021	This paper proposes a smart assistive navigation system combining ultrasonic, accelerometer, and environmental sensors to detect obstacles in multiple directions. It enhances navigation by providing multi-level alerts to visually impaired users.	The system achieves improved detection accuracy and better user safety through sensor fusion, enabling detection of front, side, and uneven surface hazards. It demonstrates reliable performance indoors and	✓ Ultrasonic Sensors ✓ Accelerometer & Environmental Sensors ✓ Microcontroller-Based Processing ✓ Audio & Vibration Alerts ✓ Multi-Sensor Fusion Algorithm
		Focuses on integrating multiple sensing modalities to overcome limitations of single-sensor systems.	Provides more stable obstacle detection even in noisy or cluttered environments.	✓ Embedded System Design ✓ Low-Power, Portable Architecture
		Designed to operate in real-time with reduced false detections and improved mobility support.	Increased user confidence due to consistent obstacle awareness.	✓ Mixed-Mode Alerts (Audio + Haptic) ✓ Rechargeable Battery Module
Research Gaps / Limitations	Despite improved sensing, the system lacks camera-based visibility, IoT connectivity, GPS/SIM-based emergency communication, and long-range detection features offered in our ESP32-CAM design.			

PROPOSED SYSTEM

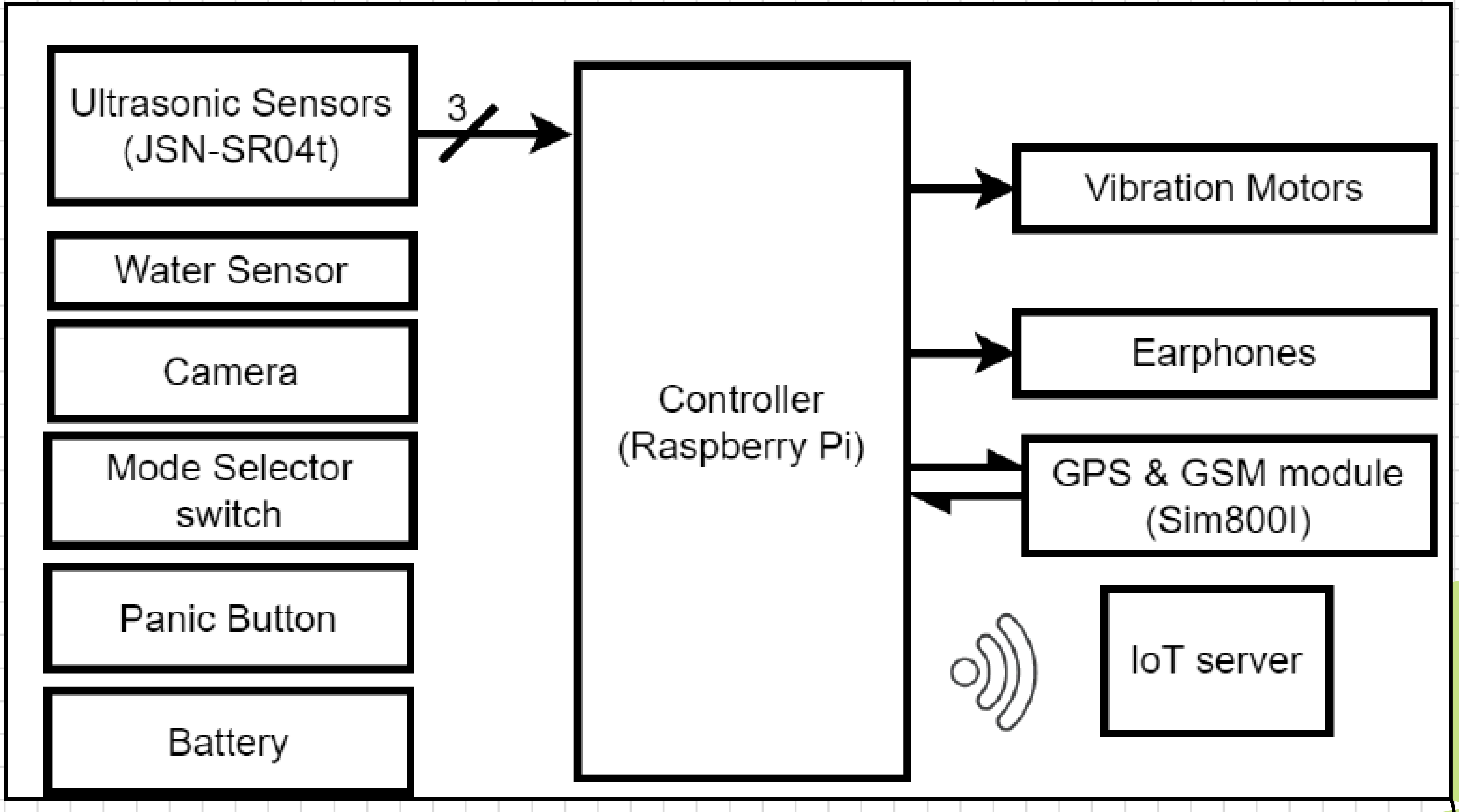
The proposed system is a smart assistive stick designed using an ESP32-CAM microcontroller to improve the safety and mobility of visually impaired individuals. It uses ultrasonic sensors to detect obstacles and a water sensor to identify puddles or wet surfaces, providing real-time alerts through vibration motors and audio feedback. An emergency panic button integrated with a GSM module allows quick communication during critical situations, while a rechargeable battery makes the system portable, efficient, and suitable for daily use.



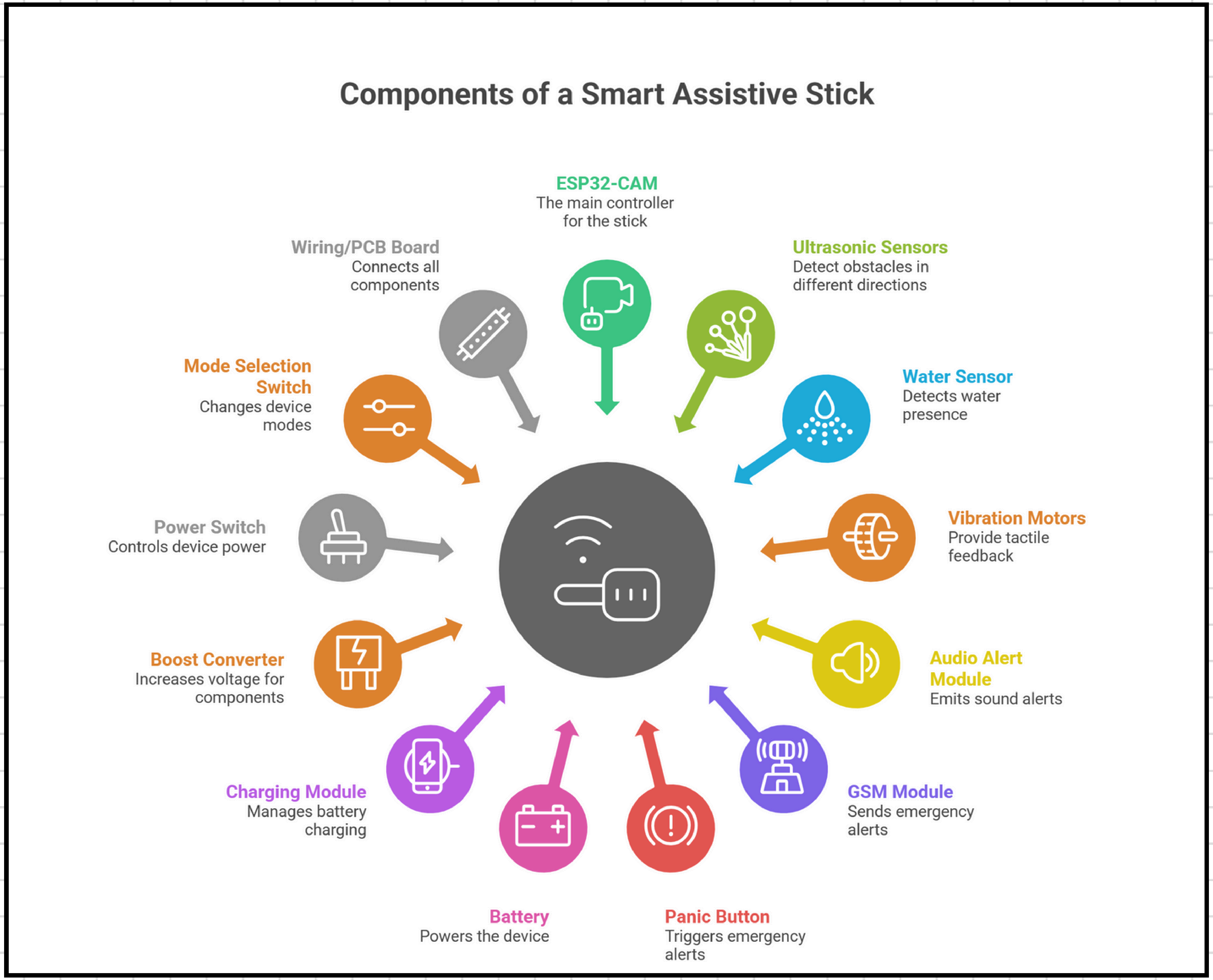


CAD Link

BLOCK DIAGRAM



COMPONENTS USED



EXPECTED OUTCOME

The proposed smart assistive stick is expected to improve safe and independent mobility for visually impaired individuals by providing early detection of obstacles and hazardous surfaces. The system will generate timely vibration and audio alerts to guide users and reduce the risk of collisions or accidents.
The integration of emergency communication through a panic button and GSM module enhances user safety in critical situations. Overall, the system aims to deliver a low-cost, reliable, and user-friendly assistive solution that increases confidence, awareness, and independence during daily navigation.

CONCLUSION

The smart assistive stick developed in this project provides an effective solution to enhance the mobility and safety of visually impaired individuals. By integrating ultrasonic sensing, water detection, real-time feedback, and emergency alert features, the system offers improved awareness of the surrounding environment. The use of an ESP32-CAM makes the design lightweight, efficient, and cost-effective, ensuring practicality for daily use. Overall, the system successfully supports independent navigation and contributes to a safer and more confident walking experience.

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